

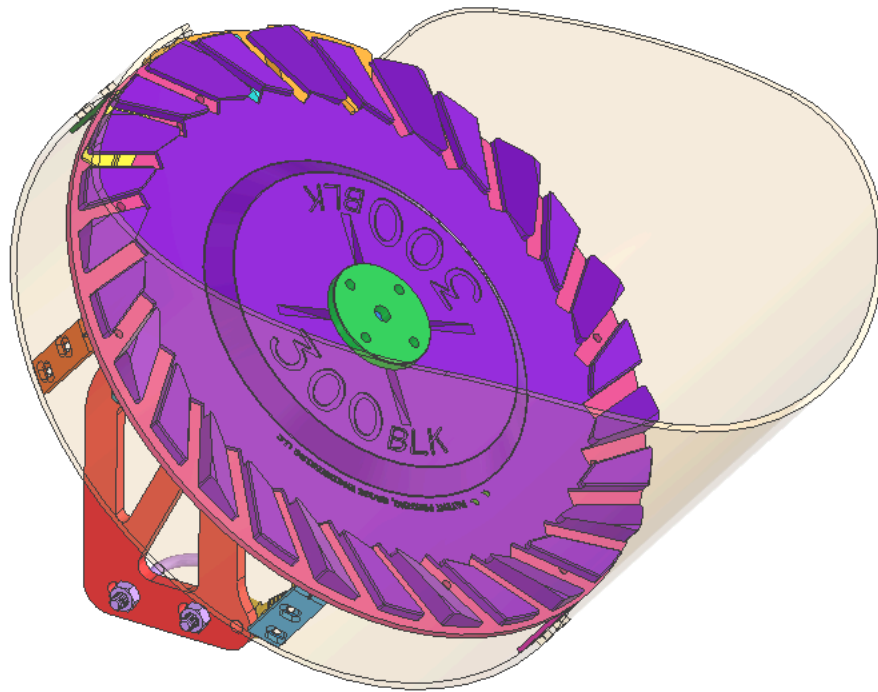
Commercial Bowl Orientation System

CBOS

Customer Manual

Rev A

Gross Engineering LLC



[Commercial Bowl Orientation and feeder]

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1. Overview

The Commercial Bowl Orientation System (CBOS) is a rotary-style bullet feeder designed for high-volume reloading operations. Unlike vibratory feeders, the CBOS uses a spinning orientation wheel to consistently present projectiles nose-up for feeding, delivering reliable, repeatable performance across a wide range of calibers.

The CBOS is designed to work with both rifle and pistol projectiles using a modular orientation plate system — swap plates to change caliber with no tools required.

The system is powered by a 12V, 2A gear motor with an adjustable speed controller, and interfaces directly with the Lee Precision pass-through die or drop die for seamless integration into any progressive press setup.

Key Features

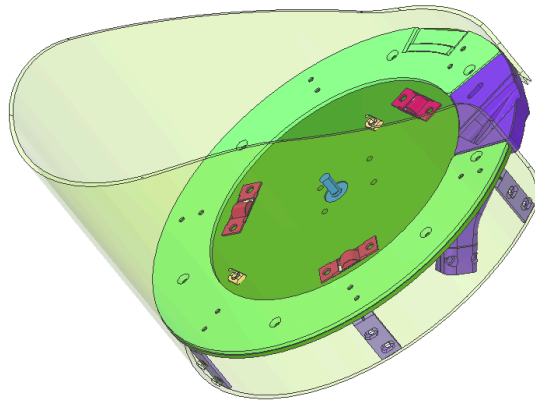
- Rotary orientation wheel for consistent bullet presentation
- Compatible with rifle and pistol calibers
- Modular orientation plate — quick caliber changes
- Lee pass-through / drop die compatible
- 12V DC gear motor with variable speed control
- Adjustable drop angle via turnbuckle
- Sensor-driven automatic operation

2. What's in the Box

Inspect all items upon receipt. Contact us immediately if anything is missing or damaged.

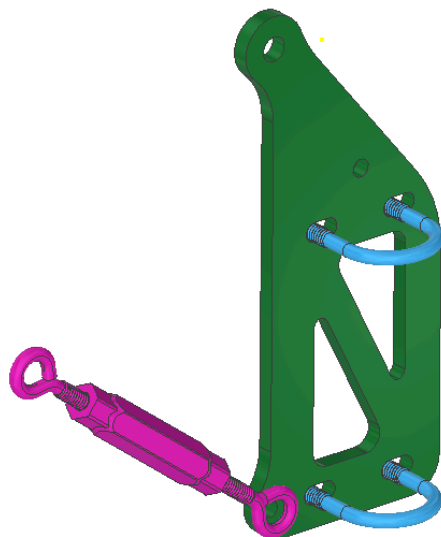
Main Bowl Assembly (pre-assembled)

- 12" orientation bowl with shroud
- Rotary orientation wheel (installed)
- Motor controller with speed dial and power switch (installed)
- Drop tube exit port



Mounting Hardware

- Pivot bracket (mounts to 1" OD pipe or extruded aluminum — customer-supplied)
- Turnbuckle (for bowl angle adjustment)



Power & Controls

- 12V / 2A power supply
- Power cable

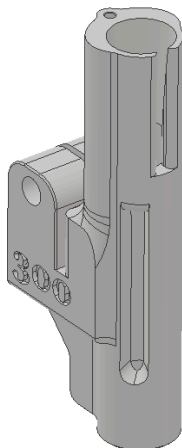
Fasteners (included)

- (2) U-bolts — pivot bracket to vertical riser
- M4 screws — motor mounting
- M6 screws — bowl and turnbuckle mounting
- #8-32 screws — general assembly

⚠ Most fasteners are pre-installed at the factory. Only the U-bolts and a small number of assembly fasteners require installation by the customer.

Included Accessories

- Bullet passage adapter (mounts on bullet drop die; includes inductive proximity sensor and spring hook for drop tube attachment)
- 3 ft. spring wire (drop tube; routes bullets from bowl to press)



You Will Also Need (Customer-Supplied)

- 1 to 1.25" OD pipe or extruded aluminum vertical riser (customer to determine height and mounting method)
- Lee Precision pass-through die or drop die (caliber-specific — see Section 10)
- Progressive reloading press
- Tools appropriate for securing your vertical riser (lag bolts, pipe clamps, etc.)

3. Before You Begin

Tools Required

Tool	Use
5/32" hex key (Allen wrench)	#8-32 fasteners
3mm hex key	M4 motor screws
5mm hex key	M6 bowl and turnbuckle screws
Adjustable wrench or pliers	U-bolt nuts
Scissors or wire cutters	Trimming spring wire drop tube
Tools for riser mounting	Lag bolts, pipe clamps, etc. — customer-dependent

Safety

- Disconnect power before making any adjustments to the motor, orientation plate, or wiring.
- The orientation wheel rotates during operation — keep fingers clear while powered.
- This system operates at 12V DC. Follow standard low-voltage wiring practices.
- Ensure your vertical riser is securely anchored before operating. The bowl assembly is top-heavy when loaded with projectiles.
- Do not exceed the recommended motor speed for your caliber — see Section 7.1.

4. Assembly

4.1 Unbox & Inspect

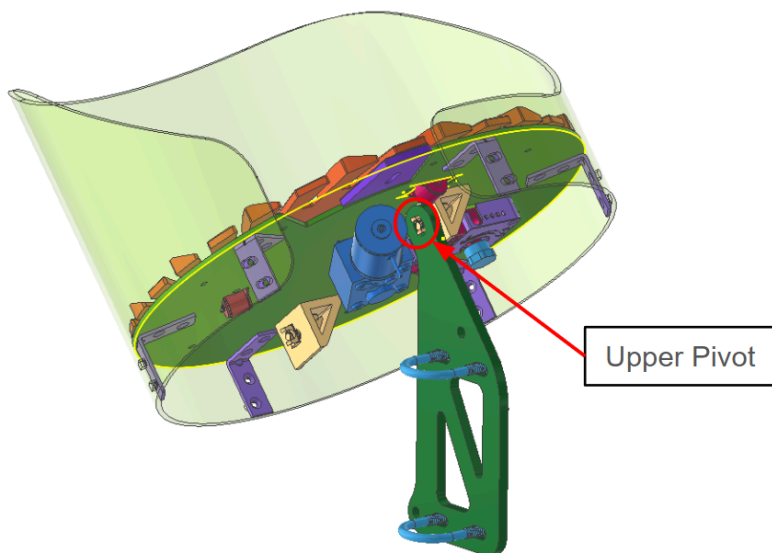
Remove all components from packaging and verify against the parts list in Section 2. Inspect for shipping damage before proceeding. Do not power on the unit until assembly is complete.

4.2 Mount the Pivot Bracket to Your Vertical Riser

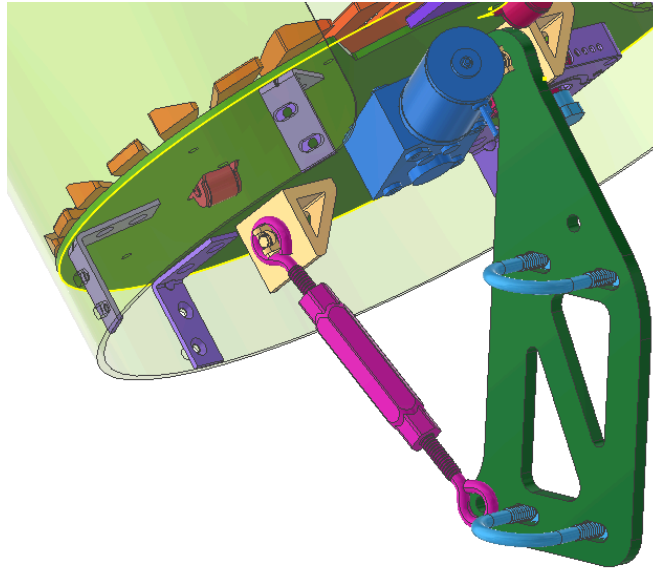
1. Position the pivot bracket at the desired height on your 1" OD pipe or extruded aluminum riser.
2. Slide the two U-bolts through the pivot bracket and around the riser.
3. Thread on the U-bolt nuts and tighten with an adjustable wrench. No rotation or play should remain.

4.3 Hang and Secure the Bowl Assembly

4. Thread the top pivot bolt through the pivot bracket and into the bowl assembly to hang it. This is your primary pivot point.
5. With the bowl hanging, attach the turnbuckle between the bottom of the bowl and the pivot bracket using the M6 bolts. Leave the turnbuckle at mid-adjustment for now — final angle is set in Section 7.4.



[Top pivot bolt installed]

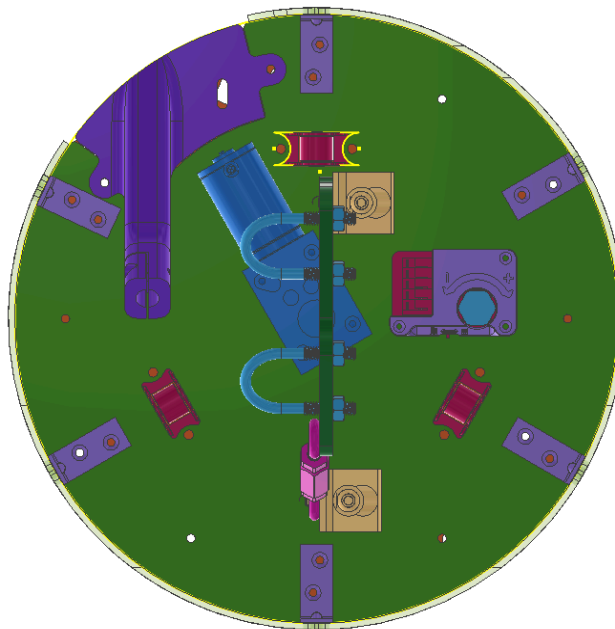


[Bottom Pivot, turnbuckle attached]

4.4 Verify Pre-Installed Components

The gear motor, orientation plate, and motor controller are pre-installed on the underside of the bowl assembly at the factory. Before proceeding to electrical connections, confirm:

- Motor is securely mounted with no movement
- Orientation plate spins freely without contacting the bowl shroud
- Controller board is secured and undamaged



[Underside showing motor and control board]

5. Electrical Connection & First Power-On

Connections

All wiring is pre-terminated at the factory. Connect leads to the clamping terminal block on the PCB by inserting each wire and tightening the clamp screw.

Motor Wiring

Wire Color	Terminal
Red	Motor +
Black	Motor –

Inductive Proximity Sensor Wiring

Wire Color	Terminal
Brown	Sensor +
Blue	Sensor –
Black	Sensor signal

Route the sensor cable from the bullet passage adapter up along your riser, away from the rotating orientation wheel, and into the terminal block.

First Power-On

6. Confirm all wires are fully seated in the terminal block and no wires are near the rotating orientation wheel.
7. Set the speed dial to minimum (fully counterclockwise).
8. Plug in the 12V power supply.
9. Slowly increase the speed dial — the orientation wheel should begin spinning smoothly.
10. If the motor does not spin or the controller does not power on, disconnect immediately and refer to Section 9 (Troubleshooting).

6. Using the Motor Controller

The CBOS controller is intentionally simple — there are no menus or parameters to configure.

Controls

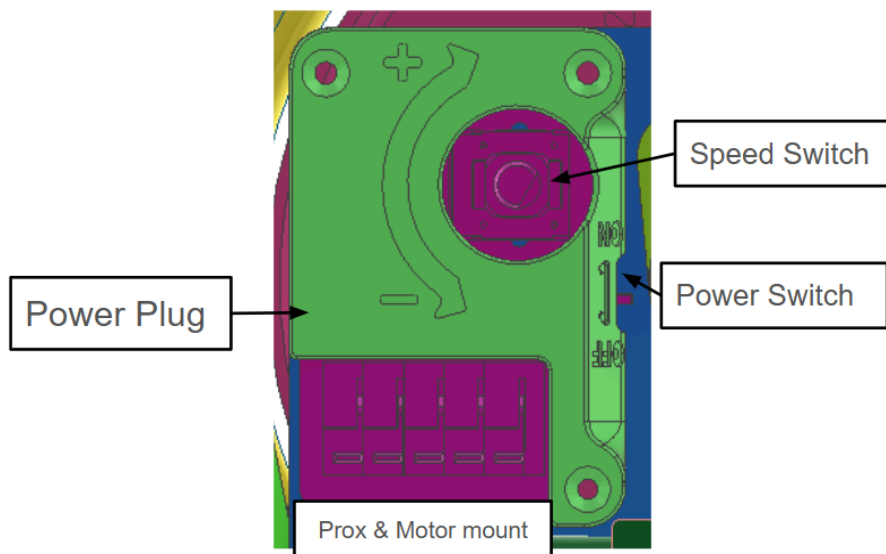
Control	Function
Power switch	Turns the system on/off
Speed dial	Adjusts orientation wheel speed — clockwise to increase, counterclockwise to decrease

How It Works

The system is sensor-driven and fully automatic. When the inductive proximity sensor detects no bullet at the bullet passage adapter, the motor runs and feeds the next projectile. When a bullet is detected and waiting, the motor stops. No user input is required during normal operation.

Speed Setting

Start with the speed dial at a low setting and increase gradually until the system feeds consistently without jamming or misfeeding. Most calibers will find an optimal speed through a short trial period — see Section 7.1 for guidance.



[Controller with switch and dial labeled]

7. Tuning & Setup

7.1 Finding Your Operating Speed

With the bowl loaded with a small number of projectiles, power on the system and set the speed dial to a low setting. Gradually increase speed until bullets are feeding consistently and presenting correctly at the bullet passage adapter.

Signs the speed is too high include bullets tumbling, stacking, or misfeeding at the exit. Back the speed down until feeding is smooth and reliable.

7.2 Installing & Trimming the Drop Tube

The spring wire runs directly from the bowl exit port to the bullet passage adapter seated in your press die. There are no intermediate fittings.

11. With your press and die in place, stretch the spring wire from the bowl exit port down to the bullet passage adapter.
12. Trim to approximate length — the spring can be elongated slightly to reach, but leave enough slack that the spring is not pulling tension on the die.
13. Hook the top end into the bowl exit port and the bottom end into the spring hook on the bullet passage adapter.

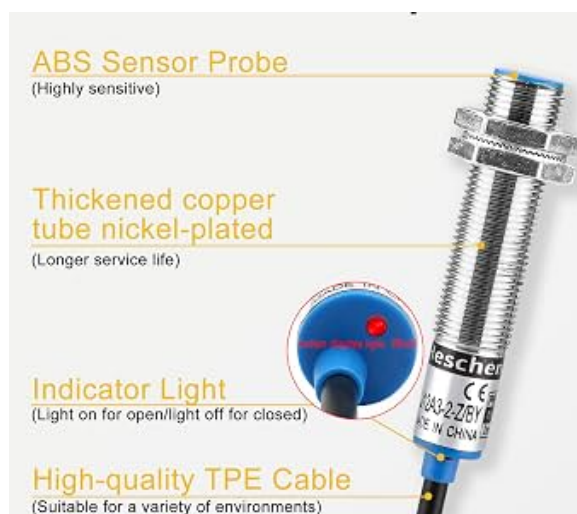
⚠ A small amount of slack is intentional — do not pull the spring taut.

7.3 Setting the Proximity Sensor Position

The inductive proximity sensor is pre-installed in the bullet passage adapter. A red indicator light on the back of the sensor shows its current state:

Indicator Light	Status
Red ON	No bullet detected — motor running
Red OFF	Bullet detected — motor stopped

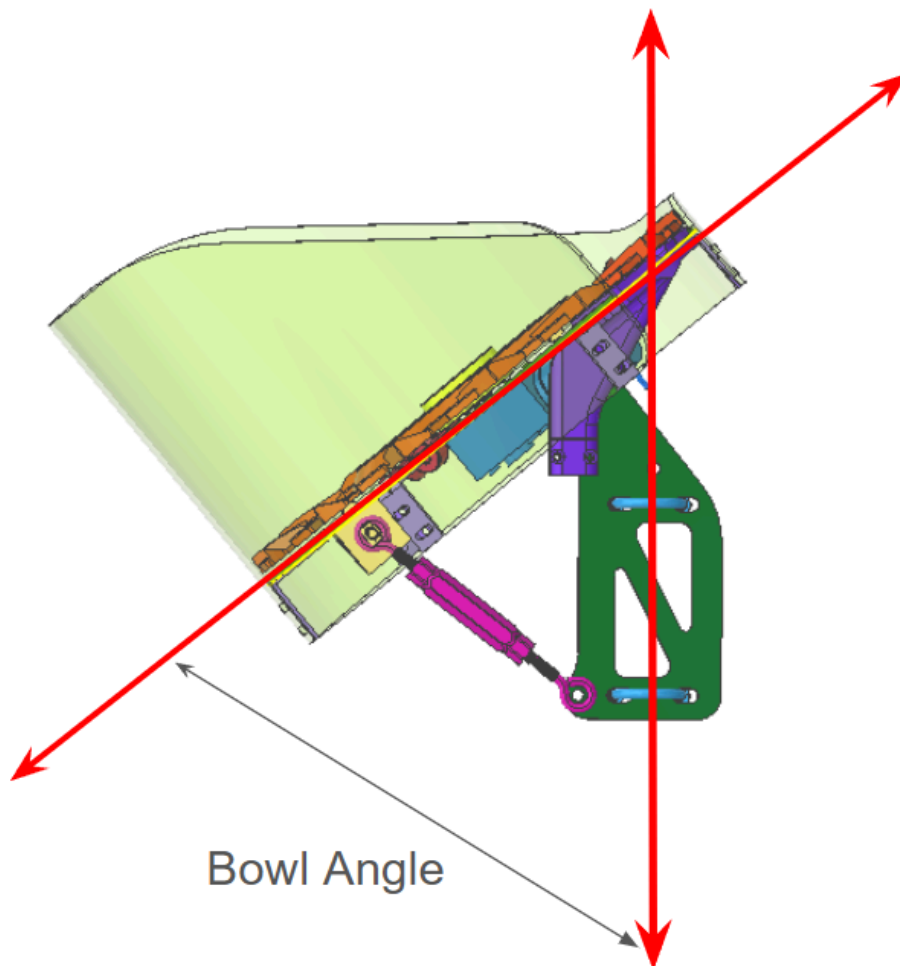
During setup, confirm the sensor is responding correctly by manually placing a bullet at the adapter and verifying the red light turns off. Remove the bullet and confirm the light returns on and the motor resumes.



[Sensor indicator light — bullet present vs. absent]

7.4 Setting the Bowl Angle (Turnbuckle)

The turnbuckle controls the angle between your vertical riser and the underside of the bowl. This angle is the primary tuning adjustment for correct bullet orientation.



How It Works

As projectiles rotate around the bowl, they reach a high point where gravity acts as the sorting mechanism. Correctly oriented bullets are captured and ride through to the drop point. Incorrectly oriented bullets tip and fall back into the bowl before reaching the drop.

Adjusting the Angle

14. Loosen the M6 bolts at the turnbuckle ends slightly.
15. Extend or shorten the turnbuckle to increase or decrease the angle between the riser and the bowl underside.
16. Retighten once adjusted and run a small batch to evaluate.

Diagnosing

Symptom	Adjustment
Both correct and wrong orientation bullets passing through to the drop	Increase the angle — correct orientation bullets need to be more firmly trapped
No bullets are passing through	Decrease the angle — the threshold is too aggressive
Cannot find a consistent sweet spot	Change the orientation plate — a different plate shifts the bullet's center of gravity, causing it to tilt more predictably at the high point

8. Normal Operation

Loading the Bowl

17. Start with a small quantity of projectiles — a handful to a cup is a good starting point.
18. Power on the system and observe feeding behavior before adding more.
19. Continue adding projectiles gradually until you reach a quantity you're comfortable running.

The bowl can hold more than you need to start — adding too many at once puts unnecessary strain on the motor and orientation wheel. Find a comfortable running quantity and maintain it.

Running

Once loaded, the system runs automatically. The motor starts and stops based on the proximity sensor — no input is required during normal operation. Monitor the bowl level periodically and top up as needed.

Motor Load & Torque

Running the bowl at high fill levels increases load on the gear motor. If you notice the motor struggling, slowing, or stalling under load:

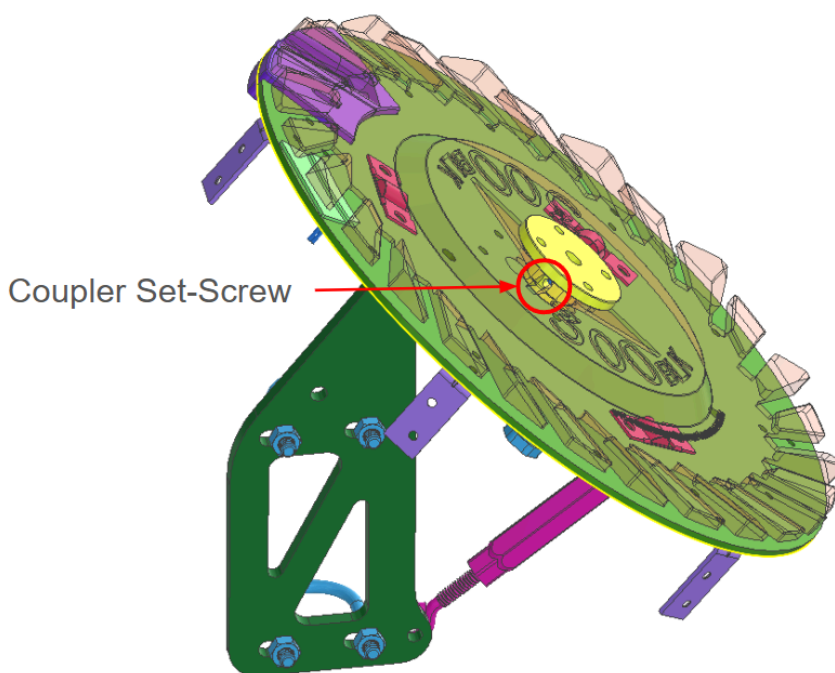
- Reduce the number of projectiles in the bowl, or
- Switch to a higher-torque motor. Note that a higher-torque motor will run at a lower speed — find a new operating speed using Section 7.1.

9. Troubleshooting

Mechanical

Motor spins but orientation plate does not move

The grub screw on the motor shaft has come loose. The motor shaft is a D-profile — locate the grub screw on the orientation plate hub, seat the plate firmly onto the flat of the shaft, and tighten the grub screw securely.



[Grub screw location on orientation plate hub]

Broken tooth on orientation plate

Replace the orientation plate. Plates are 3D printed in ASA — a replacement STL file is available from Gross Engineering LLC (see Section 10). Install the new plate and re-seat the grub screw per the above.

Spring drop tube keeps falling out

The spring fit has become loose. Use a small strip of painter's tape to shim the spring at the attachment point to add friction and hold it in place.

Electrical

Bowl does not turn on — proximity sensor red light is on

Power is reaching the board. Check that the red and black motor wires are fully seated in the terminal block. If wires are correctly seated, inspect the bowl for a physical obstruction preventing the orientation wheel from rotating and clear it before powering on again.

Bowl rotates in the wrong direction

Swap the red and black motor wires on the terminal block to reverse motor polarity and correct rotation direction.

Board has no power

Check the power supply connection. If the connection is secure and the board remains unpowered, replace the power supply or the control board. See Section 10 for sourcing.

10. Maintenance

Cleaning the Bowl Shroud

Wipe the bowl and shroud with a clean dry cloth after each use. Lead and copper fouling from projectiles will accumulate over time — use a mild solvent on a cloth to remove buildup. Do not spray solvents directly onto the motor or PCB.

Replacement Parts

All components of the CBOS are sourceable. Below are specifications and sources for every serviceable part.

Gear Motor

- BRINGSMART 70kg-cm 12V DC gear motor
- [Available on Amazon \(B0774HGZGX\)](#)
- When replacing, a higher-torque / lower-speed variant may be substituted — see Section 8 for torque guidance

Inductive Proximity Sensor

- Heschen PR12-4DC, M12 cylindrical, NPN NC, 4mm sensing distance
- [Available on Amazon \(B0816F62W6\)](#)
- Wire colors: Brown (+), Blue (–), Black (signal)

Orientation Plate

- 3D printed in ASA
- STL files available from Gross Engineering LLC — see GitHub repository [link TBD]

PCB / Control Board

- Available direct from Gross Engineering LLC — see GitHub repository [link TBD]

Fasteners

- #8-32 — standard hardware store
- M4 — standard metric hardware
- M6 — standard metric hardware
- U-bolts (1" ID) — standard hardware store or plumbing supply

Lee Inline Bullet Feed Die — Caliber Reference

The Lee Precision Inline Bullet Feed Die is required for press interface and is ordered separately by caliber.

[Lee Precision — Inline Bullet Feed Dies](#)

Caliber Group	Lee SKU	Bullet Diameter Range	Compatible Cartridges
22 CAL	91998	.220—.225"	22 Hornet to 222 Rem Mag
24 CAL	92025	.238—.245"	6mm, 243 Win
26 CAL	92000	.250—.277"	6.5 Creedmoor and similar
28 CAL	92214	.275—.290"	270 Win to 7mm Rem Mag
30 CAL	91999	.250—.320"	30 M1 to 30-06
35 CAL	91995	.320—.360"	380 Auto, 9mm to 357 Mag
40 CAL	91996	.390—.405"	40 S&W
44 CAL	92071	.425—.435"	44 Mag, 44 Special
45 CAL	91997	.435—.460"	45 ACP to 45 Colt